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MPhys Project Report

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Cracking the Cosmos

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Abstract: Vainshtein screening is a non-linear mechanism that suppresses new long-range forces within a characteristic Vainshtein radius, allowing modifications of gravity to avoid local tests. In this report, we study Vainshtein screening in an infrared Galileon effective field theory, focusing on the role of higher-dimension operators. Using state-of-the-art numerical methods implemented using the φenics software, we solve the equations of motion for this theory. We find that in the Vainshtein regime, higher-dimension operators can dominate counter to the expectation for an effective field theory.

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1 Introduction

Einstein's theory of General Relativity (GR), developed at the beginning of the 20th century, fundamentally changed our understanding of gravity. Rather than a force acting at a distance, gravity is described as a manifestation of the curvature of spacetime. The dynamics are captured by Einstein's field equations,

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \ , \tag{1.1}$$

where $G_{\mu\nu}$ is the Einstein tensor, G is Newton's gravitational constant, c is the speed of light, and $T_{\mu\nu}$ is the energy-momentum tensor. The elegance of this theory is that it emerged from concepts such as the Einstein Equivalence Principle (EEP) and the requirement of general covariance. From Einstein's 'happiest thought' to a framework capable of explaining phenomena ranging from black holes to an apple falling from a tree, GR is undoubtedly one of the great triumphs of theoretical physics. Indeed, Dirac remarked that it is 'probably the greatest scientific discovery that was ever made' [1].

Einstein outlined three tests for GR when he proposed it: the anomalous perihelion precession of Mercury, the deflection of light by the Sun, and gravitational redshift [2]. The discrepancy in Mercury's orbit was already known, so in that sense it was a *retrodiction*. However, as argued by Weinberg [3], this may have represented a stronger confirmation of the theory than the prediction of light deflection, subsequently verified by Eddington [4, 5].

More recently, significantly more stringent experimental constraints have been placed on deviations from the predictions of GR. For instance, measurements of the gravitational deflection of light now achieve a precision approximately three orders of magnitude better than Eddington's original eclipse observations [6, 7]. Furthermore, phenomena such as the Shapiro time delay [8, 9] provided complementary probes of the same underlying gravitational dynamics. For an excellent review of experimental tests of gravity, see [10].

Despite the remarkable empirical success of GR, a wide range of alternative theories of gravity have continued to be proposed, many of which aim to recover GR in an appropriate limit (much as Einstein's theory reduces to Newtonian gravity). The number of such models is in fact large enough that we do not endeavour to produce a comprehensive list here. Some examples include higher-dimensional theories of gravity, such as Kaluza-Klein models [11, 12], braneworld scenarios [13, 49], scalar-tensor theories such as Brans-Dicke gravity [14] and $f(R)$ theories. As outlined in [7], the abundance of alternative theories of gravity, together with advances in experimental precision, necessitated a systematic framework for comparing theory with observation. This role is fulfilled by the Parameterised Post-Newtonian (PPN) formalism [15], which characterises weak gravity, allowing relevant experimental bounds to be

applied to a broad range of theories without requiring detailed knowledge of their specific subtleties.

Many theories of *modified gravity* arise from theoretical considerations, exploring which theories are consistent with a given set of underlying principles and assumptions, after which their phenomenological consequences can be studied. However there also exist strong observational and theoretical motivations to depart from Einstein's theory.

A prominent example is the observed accelerated expansion of the Universe. Measurements from, for instance, the Cosmic Microwave Background and type Ia supernovae are consistent with a Λ-cold dark matter (ΛCDM) cosmological model [16]. The observed value of the cosmological constant (CC) is remarkably small in Planck units, $\Lambda_{\rm obs} \sim 10^{-120} M_{\rm Pl}^4$. If we interpret this as a vacuum energy density, then a heuristic estimate can be obtained by summing the zero-point energies of Standard Model (SM) fields up to an ultraviolet cutoff $\Lambda_{\rm UV}$ [17],

$$\langle \rho \rangle \sim \int_0^{\Lambda_{\rm UV}} {\rm d}k\, k^2 \sqrt{k^2 + m^2} \sim \Lambda_{\rm UV}^4 \,, \tag{1.2}$$

where $\Lambda_{\rm UV}$ characterises the scale at which the effective field theory (EFT) is expected to break down. Taking a conservative cutoff of order $\Lambda_{\rm UV} \sim 1$ TeV, consistent with tests up to the weak scale [18], we find $\Lambda_{\rm theory} \sim 10^{-60} M_{\rm Pl}^4$. The *cosmological constant problem* is not the smallness of the observed value of Λ per se, but the enormous discrepancy between this value and the vacuum energy contributions expected from quantum field theory.[1]

The modification of gravity considered in this paper is the introduction of an additional scalar field. Any such modification must reproduce GR at well-tested scales, such as within the solar system. This could be satisfied by assuming extremely weak couplings, such that deviations from GR lie below current experimental sensitivities, or — more interestingly — by invoking 'screening mechanisms' that dynamically suppress observable effects of the scalar field in certain regions [7]. These are particularly appealing as they allow for significant modifications on cosmological scales whilst preserving agreement with local tests of gravity.

Several screening mechanisms have been proposed in the literature. One example is the chameleon mechanism [19], in which the mass of the scalar field becomes large in high-density environments, suppressing its effects locally. In this work, however, we focus on the Vainshtein mechanism [20], which operates through nonlinear derivative interactions of the scalar field.

In particular, we consider a Galileon [21] EFT of the form presented in [22]. The regime of validity of this EFT is investigated using φenics, the numerical software developed and presented in that work.

[1] Physicists get a thrill from finding and naming problems; see the flatness problem, horizon problem, hierarchy problem, measurement problem ...

The report begins in subsection 2.1 with a brief review of general relativity formulated in terms of an action, before moving on to the explore the Vainshtein mechanism in subsection 2.2. The specific model we consider in this report is that of a massive Galileon, so the landscape of Galileon theories is introduced in subsection 2.3. The numerical code used in the analysis, φenics, is described in subsection 3.2, after the model that it implements is introduced in subsection 3.1. Results concerning the model phenomenology and EFT framework are presented in section 4, with section 5 discussing the conclusions drawn.

2 Theory

2.1 General Relativity

In section 1, we introduced Einstein's field equations (Equation 1.1); here we define General Relativity to be the theory described by the Einstein-Hilbert action

$$S_{\mathrm{EH}} = \frac{1}{16\pi G} \int R \, d^4x \sqrt{-g} \, , \tag{2.1}$$

where $g_{\mu\nu}$ is the metric, $g \equiv \det g_{\mu\nu}$, $\sqrt{-g}\, d^4x \equiv \sqrt{-g}\, \mathrm{d}^0x \wedge \cdots \wedge \mathrm{d}^3x$ is the invariant volume element [23], G is Newton's gravitational constant and R is the Ricci scalar. It can be shown [24, 25] that $\delta S_{\mathrm{EH}} = 0$ under variation δg leads to the vacuum Einstein equation,

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 0 \, , \tag{2.2}$$

where $R_{\mu\nu}$ is the Ricci tensor and $G_{\mu\nu}$ is the Einstein tensor.[2] If one introduces a matter action S_{M}, the energy-momentum tensor $T^{\mu\nu}$ can be defined in terms of the change δS_{M} under δg. Then analogously to above but with $\delta(S_{\mathrm{EH}} + S_{\mathrm{M}}) = 0$, the Einstein equation is obtained

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} \, . \tag{2.3}$$

Note that the normalisation of S_{EH} is chosen to reproduce the Newtonian limit; it is also common to set $c = 1$.

Many modifications to General Relativity, although not all, can be expressed as modifications of the Einstein-Hilbert action. While GR has been remarkably successful, it is generally considered to be an EFT valid only below the Planck scale. At higher energies, additional higher-dimensional operators are expected to become relevant, as their suppression by powers of the Planck mass M_{Pl} breaks down [23]. This is analogous to Fermi's theory of weak interactions, which provides an accurate description at energies below the electroweak scale but beyond which must be replaced by the full electroweak theory where the W and Z bosons become relevant.

[2] The results in this section are only stated; for more details on the process, consult the given references or your favourite book on GR.

2.2 Vainshtein mechanism

The Vainshtein mechanism was originally proposed in [20] as a solution to the van Dam-Veltman-Zakharov (vDVZ) discontinuity [48] that arises in Fierz-Pauli (FP) massive gravity [26]. This is a linear theory developed in the 1930s, corresponding to linearised GR with the unique ghost-free mass term

$$\mathcal{L}_{\mathrm{FP,mass}} = -\frac{1}{4} M_{\mathrm{Pl}}^2 m^2 (h_{\mu\nu} h^{\mu\nu} - h^2)\,, \tag{2.4}$$

where $h_{\mu\nu}$ is the metric perturbation about flat spacetime, $h^2 \equiv (\eta^{\mu\nu} h_{\mu\nu})^2$, m is a mass parameter, and the relative minus between terms ensures no ghost [27]. Here, a *ghost* refers to a degree of freedom whose excitations carry negative energy, most commonly arising from a kinetic term with the wrong sign. Such states render the vacuum unstable and, if present below the cutoff of an effective field theory, signal a fundamental inconsistency [18].

The vDVZ discontinuity arises from the fact that in the $m \to 0$ limit, linearised GR is not recovered. Specifically, the propagators of the massless theory and the massive theory do not agree in this limit, as explained in [28]. As a consequence, the FP theory suffers from the pathology that it yields incorrect physical predictions (e.g. bending of light) even for arbitrarily small graviton mass.[3]

Vainshtein showed that the vDVZ discontinuity is in fact an artefact of working at the linear level. Near sufficiently massive sources, non-linear interactions become important for small graviton masses [20]. Within what is now known as the *Vainshtein radius* r_{V}, the helicity-0 mode of the massive graviton (which can be described as an effective scalar degree of freedom (dof)) has its gradients suppressed due to non-linear derivative self-interactions. As a result, the scalar-mediated force is screened, allowing the theory to recover the phenomenology of GR.

As an aside, the FP theory is troublesome even with the Vainshtein mechanism due to the Boulware-Deser ghost [29] which appears when including non-linearities. Recently, progress has been made with de Rham-Gabadaze-Tolley (dRGT) gravity, a theory of massive gravity in 4D which is free of ghosts in the decoupling limit [30].

To sketch how this works, we consider a scalar field ϕ conformally coupled to matter [31]

$$\mathcal{L} = -\frac{1}{2} Z^{\mu\nu}(\phi) \partial_\mu \phi \partial_\nu \phi - V(\phi) + A(\phi) T\,, \tag{2.5}$$

where $Z_{\mu\nu}$ is an effective kinetic coefficient, V is a potential, and A describes the coupling to the trace of the energy-momentum tensor $T = T^\mu_\mu$. If the kinetic term is of the form

$$Z_{\mu\nu} \sim g_{\mu\nu} + \frac{1}{\Lambda^3} \partial_\mu \partial_\nu \phi + \frac{1}{\Lambda^6} \left(\partial_\mu \partial_\nu \phi\right)^2 + \dots\,, \tag{2.6}$$

[3] Frequently a theory is described as suffering from a 'pathology'. This essentially means that the theory has some fatal flaw, e.g. ghosts, internal inconsistency, unphysical predictions.

then we see that for high energies, i.e. $Z(\bar{\phi}) \gg 1$ or $\partial^2\bar{\phi} \gg \Lambda^3$ where $\bar{\phi}$ is the background configuration, higher-order interactions become important and it is no longer true that $Z_{\mu\nu} \sim g_{\mu\nu}$. As outlined in [18], if the field is expressed as a background plus fluctuation φ, then in the case of a point source, $\rho = \mathcal{M}\delta^{(3)}(\mathbf{x})$, we retrieve a static potential of the form

$$V(r) \sim -\frac{1}{Z(\bar{\phi})}\frac{\exp\left(-\frac{m(\bar{\phi})r}{\sqrt{Z(\bar{\phi})}}\right)}{4\pi r}\mathcal{M}\,. \tag{2.7}$$

The force $F_\varphi \sim 1/r^2$ can thus be suppressed when $Z(\bar{\phi})$ becomes large, as explained previously. Galileon theories can provide a realisation of this mechanism through their non-linear interactions.

2.3 Galileons

Before explicitly demonstrating the Vainshtein mechanism, we introduce the theoretical framework in which it arises in this work. Galileon theories were originally identified in the decoupling limit ($M_{\rm Pl}, M_5 \to \infty$) of the Dvali-Gabadaze-Porrati (DGP) braneworld model [13] mentioned in section 1 [31].

Since their original formulation, galileons have appeared in a variety of settings and are now studied in their own right. Notable developments include extension to curved backgrounds [32], covariant formulations [33], and their interpretation as EFTs [34]. Galileon interactions have also been explored in cosmology, for instance in the context of inflation [35] [36–38].

As outlined in [21], the defining feature of galileon theories is an internal symmetry of the scalar field π, under which the Lagrangian $\mathcal{L}_\pi$ is invariant:

$$\pi(x) \to \pi(x) + b_\mu x^\mu + c\,, \tag{2.8}$$

where c is a constant shift and b_μ characterises a constant gradient shift $\partial_\mu\pi \to \partial_\mu\pi + b_\mu$.[4] Hopefully, this should look reminiscent of the Galilean symmetry from classical mechanics $x^i \to x^i + v^i t$ or $\dot{x} \to \dot{x} + v$, hence the term *galileon*. We also require the equation of motion (eom) to be invariant under Equation 2.8, i.e. that any π appearing must be acted on by at least two derivatives. Finally, to avoid ghosts the π eom must be second order (see Ostrogradsky instability [39]), or of the form

$$\frac{\delta\mathcal{L}_\pi}{\delta\pi} = F(\partial_\mu\partial_\nu\pi)\,, \tag{2.9}$$

where F is some non-linear Lorentz invariant function. It can be shown that, at each order in π, there exists a unique interaction term (up to total derivatives)

[4] Pedantically, we mean the Lagrangian *density*, with the actual Lagrangian L being the integral of $\mathcal{L}$ over a Cauchy surface $\int_\Sigma$.

that satisfies the galileon symmetry while yielding second-order equations of motion. Therefore, in d dimensions there are exactly $(d+1)$ galileon terms. Each term takes the schematic form

$$\mathcal{L}_n \sim (\partial^2\pi)^{n-2}\partial\pi\partial\pi\,, \tag{2.10}$$

with contractions chosen such that higher-derivative contributions to the eom cancel. In four-dimensions, the Galileo-invariant Lagrangians are [18]

$$\begin{aligned}
\mathcal{L}_1 &= \pi\,,\\
\mathcal{L}_2 &= \frac{1}{2}(\partial\pi)^2\,,\\
\mathcal{L}_3 &= \frac{1}{2}\Box\pi(\partial\pi)^2\,,\\
\mathcal{L}_4 &= \frac{1}{4}(\partial\pi)^2\left((\Box\pi)^2-(\partial_\mu\partial_\nu\pi)^2\right)\,,\\
\mathcal{L}_5 &= \frac{1}{3}(\partial\pi)^2\left((\Box\pi)^3+2(\partial_\mu\partial_\nu\pi)^3-3\Box\pi(\partial_\mu\partial_\nu\pi)^2\right)\,,
\end{aligned} \tag{2.11}$$

where $\Box\pi \equiv \partial^\mu\partial_\mu\pi$ is the d'Alembertian and we employ the shorthand $(\partial\pi)^2 \equiv \partial_\mu\pi\partial^\mu\pi$. For completeness, we note that it is also common in the literature to see the following notation: eom $\mathcal{E} \equiv \delta\mathcal{L}/\delta\pi$; $\Pi^\mu{}_\nu \equiv \partial^\mu\partial_\nu\pi$; $[\cdot]$ to denote the trace operator; $(\partial_\mu\partial_\nu\pi)^n \equiv [\Pi^n]$.

The general Lagrangian is then simply the linear combination [21],

$$\mathcal{L}_\pi = \sum_{i=1}^{5} c_i\mathcal{L}_i\,, \tag{2.12}$$

where c_i are coefficients.

Before continuing, we mention that Horndeski [40] worked out the Lagrangian for the most general scalar-tensor theory in four-dimensions that yields second-order field equations.[5] This is shown to be equivalent to the Galileon theory mentioned previously in [37].

To analytically see the Vainshtein mechanism, let's consider the cubic Galileon theory

$$\mathcal{L} = -3(\partial\pi)^2 - \frac{1}{\Lambda^3}(\partial\pi)^2\Box\pi + \frac{1}{M_{\rm Pl}}\pi T\,, \tag{2.13}$$

where Λ is the strong-coupling scale of the theory and $T = T^\mu_\mu$ is the trace of the energy-momentum tensor, and normalisation is chosen for conistency with [18]. Notice the presence of the $\mathcal{L}_3$ term from Equation 2.11 (hence the term 'cubic') and the coupling of the scalar field π to the energy-momentum. In the language of Equation 2.5, $V(\pi) = 0$ and $A(\pi) = \pi/M_{\rm Pl}$. Expanding about a background, the effective kinetic matrix Z corresponds to the first two terms of Equation 2.6.

[5] For an interesting historical review, see [41].

The equation of motion is then

$$6\Box\pi + \frac{2}{\Lambda^3}\Big((\Box\pi)^2 - (\partial_\mu\partial_\nu\pi)^2\Big) = -\frac{1}{M_{\rm Pl}}T\,. \tag{2.14}$$

For a static point source of mass M and assuming spherical symmetry, we consider the positive branch of the background solution π_0 [31],

$$\pi_0'(r) = \frac{3}{4}r\Lambda^3\left(-1+\sqrt{1+\frac{1}{9\pi}\left(\frac{r_{\rm V}}{r}\right)^3}\right)\,, \tag{2.15}$$

where $'$ denotes differentiation with respect to (wrt) the radial coordinate r and we define the *Vainshtein radius* to be

$$r_{\rm V} \equiv \frac{1}{\Lambda}\left(\frac{M}{M_{\rm Pl}}\right)^{1/3}\,. \tag{2.16}$$

This radius can be identified as the scale where the behaviour of solutions switches:

- $r \gg r_{\rm V}$: $\quad \pi_0' \sim 1/r^2$
- $r \ll r_{\rm V}$: $\quad \pi_0' \sim 1/\sqrt{r}$

We see that the force is thus suppressed compared to the standard gravitational force below the Vainshtein radius.

The analysis in this section was intended to be a brief sketch, and closely followed that carried out in [18, 31]. See both for a more in-depth discussion of topics such as quantum corrections and non-renormalisation.

While the cubic Galileon provides an instructive analytic illustration of Vainshtein screening, models of any real complexity require numerical methods to understand. In section 3, we introduce the numerical framework used in this work.

3 φenics

In this section we give an overview of the φenics software [42] and the infra-red (IR) model studied in this paper.

3.1 IR theory

The model we consider in this paper originates as the ultraviolet (UV) completion of a massive Galileon theory originally presented in [43]. By integrating out a heavy scalar field, this construction gives rise to a non-linear IR effective field theory, which is the theory implemented using φenics. The UV action is

$$S_{\rm UV}[\pi,H] = \int d^4x\Big(-\frac{1}{2}(\partial\pi)^2 - \frac{1}{2}(\partial H)^2 - \alpha H\Box\pi - \frac{1}{2}m^2\pi^2 - \frac{1}{2}M^2H^2 - \frac{\lambda}{4!}H^4\Big)\,, \tag{3.1}$$

where H is a heavy field with $M_H \gg m$. This hierarchy of masses ensures that H can be integrated out to achieve the low-energy description. Details of this process are given in [44], and we adapt our notation now to align with theirs.[6] Generalising to arbitrary powers of n for the heavy field, Burrage et al. write the action as

$$S[\pi, H] = \int d^4x \Big(-\frac{1}{2}(\partial\pi)^2 - \frac{1}{2}(\partial H)^2 - \frac{1}{2}m^2\pi^2 - \frac{1}{2}M^2H^2 - \alpha H\Box\pi - \frac{\lambda H^{n+1}}{(n+1)!\mu^{n-3}} + \frac{\pi T}{M_{\rm Pl}} \Big) . \quad (3.2)$$

They then make the assumption that $\Box \ll M^2$, and substitute an expression for H in terms of π back into the action to get a low energy action,

$$\tilde{S}[\pi] = \int d^4x \left(-\frac{1}{2}(\partial\pi)^2 - \frac{1}{2}m^2\pi^2 + \frac{(-1)^n\lambda\alpha^{n+1}(\Box\pi)^{n+1}}{(n+1)!M^{2(n+1)}\mu^{n-3}} + \frac{\pi T}{M_{\rm Pl}} \right) . \quad (3.3)$$

Identifying $M^{2(n+1)}\mu^{n-3} \sim \Lambda^{3n-1}$ recovers the action

$$S[\pi] = \int d^4x \left(-\frac{1}{2}(\partial\pi)^2 - \frac{1}{2}m^2\pi^2 + \frac{\epsilon}{n+1}\frac{(\Box\pi)^{n+1}}{\Lambda^{3n-1}} + \frac{\pi T}{M_{\rm Pl}} \right) . \quad (3.4)$$

This derivation is beyond the scope of the report, and we include an outline of it simply for completeness. Of interest to us is that this action gives rise to the equation of motion

$$\Box\pi - m_\pi^2\pi - \epsilon\frac{\Box\left((\Box\pi)^n\right)}{\Lambda^{3n-1}} = \frac{\rho}{M_{\rm Pl}}. \quad (3.5)$$

Equation 3.5 is a low-energy (IR) theory describing a massive scalar field π with a cutoff for the EFT of Λ. As a sanity check, we note that Λ has dimensions of mass, ensuring that the interaction term is correctly suppressed by the EFT cutoff according to dimensional power counting. The essence of this project is contained in the non-linear term, whose strength is controlled by ϵ, which gives rise to Vainshtein screening.

In the φenics paper and this report, we assume a static and spherically symmetric field, so

$$\Box \to \nabla_r^2 \equiv \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) , \quad (3.6)$$

and the eom can be expressed as

$$\nabla_r^2\pi - m_\pi^2\pi - \epsilon\frac{\nabla_r^2\left((\nabla_r^2\pi)^n\right)}{\Lambda^{3n-1}} = \frac{\rho}{M_{\rm Pl}} . \quad (3.7)$$

The boundary conditions we apply to this theory are [22]

$$\pi(\infty) = 0, \quad \frac{\partial}{\partial r}\pi(0) = 0, \quad [(\nabla_r^2\pi)^n](\infty) = 0, \quad \frac{\partial}{\partial r}[(\nabla_r^2\pi)^n](0) < \infty . \quad (3.8)$$

[6] Also note that ϵ, Λ in [44] and [22] are unfortunately not the same.

Due to the presence of nested Laplacians and non-linear powers of derivatives of the field, analytic solutions are not available in general, motivating a numerical approach. We next discuss how we approach solving this highly non-linear equation.

3.2 Numerical solution of equations of motion

To obtain numerical solutions to the IR eom boundary value problem Equation 3.7, we employ the approach implemented in the φenics package [42]. This combines a Newton-Raphson iteration with a finite element discretisation of the spatial domain.

To begin, we write the equation of motion as a functional equation

$$\mathcal{F}[\pi] = 0\,, \tag{3.9}$$

where $\mathcal{F}$ is a non-linear differential operator. Newton's method proceeds by iteratively improving an approximate solution $\pi^{(k)}$ by solving a linearised equation for the correction to the current iterate, $\delta\pi^{(k)} \equiv \pi_{\text{true}} - \pi^{(k)}$. This is then used to find an expression for $\pi^{(k+1)}$ in terms of $\pi^{(k)}$ [22]. The aim is to move closer to the true solution (within some tolerance) after repeated iterations.

Since the problem is defined on an infinite radial domain, we truncate the spatial interval to $r \in [0, r_{\text{max}}]$, with r_{max} chosen large compared to the Compton wavelength of the field so that the boundary condition $\pi(r_{\text{max}}) = 0$ is a good approximation. The truncated domain is then discretised into a finite number of elements. Within each finite element, the field is expanded in a local polynomial basis, and continuity across neighbouring elements is enforced to obtain a globally well-defined field configuration. This reduces the continuum problem to a system of coupled algebraic equations for the expansion coefficients.

The procedure described above constitutes the finite element method (FEM). In one spatial dimension this approach may appear excessive; however, the FEM is particularly powerful in higher dimensions, where complex geometries can be efficiently represented using finite element cells. For a more in-depth mathematical treatment, see [45], and for details of numerical implementation, see [46]. While we make use of the state-of-the-art φenics software tailored to our screening problem, the underlying finite element machinery (and the numerical implementation more generally) is provided by the `FEniCS` library [47].[7]

Rather than enforcing the equations of motion pointwise (*strong* form), the problem is formulated in *weak* form by integrating the differential operator against a set of test functions, requiring all inner products to vanish. This has the advantage of reducing the order of derivatives appearing in the equations and is particularly well-suited to finite element methods.

Convergence is assessed using both the residual of the equations of motion and the change in the solution between successive iterations. The specific implementation

[7] More information on `FEniCS`, including documentation and source code, can be found here.

Parameter	Value
Scalar mass m_π	$10^{-56} M_{\rm Pl}$
Non-linearity strength ϵ	3×10^{-3}
Cutoff scale Λ	$10^{-46} M_{\rm Pl}$
Operator power n	3
Source mass M_S	$10^{28} M_{\rm Pl}$
Source radius r_S	$7 \times 10^{49} M_{\rm Pl}^{-1}$

Table 1: Parameter values used for the $n = 3$ IR model.

details, including basis choice, mesh construction and numerical integration, are described in detail in [22].[8]

One implementation detail we do mention here however is the introduction of auxiliary variables to write the eom as coupled system of lower-order equations

$$\begin{pmatrix} Y - m_\pi^2 \pi - \epsilon \nabla_r^2 W \\ Y \\ W \end{pmatrix} = \begin{pmatrix} \frac{\rho}{M_P} \\ \nabla_r^2 \pi \\ Y^n \end{pmatrix} , \tag{3.10}$$

where $Y \equiv \nabla_r^2 \pi$ and $W \equiv Y^n$.

4 Results

4.1 $n = 3$

We first consider the $n = 3$ case of our infra-red theory, which corresponds to the model solved in [22], and provides a useful benchmark for understanding the screening behaviour of the model. For concreteness, the equation of motion is

$$\nabla_r^2 \pi - m_\pi^2 \pi - \epsilon \frac{\nabla_r^2 \left((\nabla_r^2 \pi)^3 \right)}{\Lambda^8} = \frac{\rho}{M_{\rm Pl}} , \tag{4.1}$$

where ∇_r^2 denotes the radial Laplacian assuming spherical symmetry. The parameter values used in this model are summarised in Table 1. Note that these differ from those of the φenics paper, although the matter source ρ has the same profile of a smoothed top-hat.

In Figure 1, we present a plot of the π profile and field gradient $\nabla_r \pi$ across the numerical box that was simulated. Around the $10^3\, \hat{r}$ region, where the gradient reaches its maximum absolute value, is where the Vainshtein radius $r_{\rm V}$ lies. Within the Vainshtein radius, non-linear derivative interactions dominate, leading to a suppressed radial gradient and a slowly varying field profile, which appears approximately flat on the log-log scale. Outside the Vainshtein radius however, we see there

[8] The documentation for φenics can be found here.

is a steeper constant gradient which corresponds to the Yukawa decay expected of a massive scalar field, $\sim e^{-m_\pi r}/r$. We notice that beyond the Compton wavelength m_π^{-1} of π, the field decays exponentially which is to be expected since $m_\pi r > 1$.

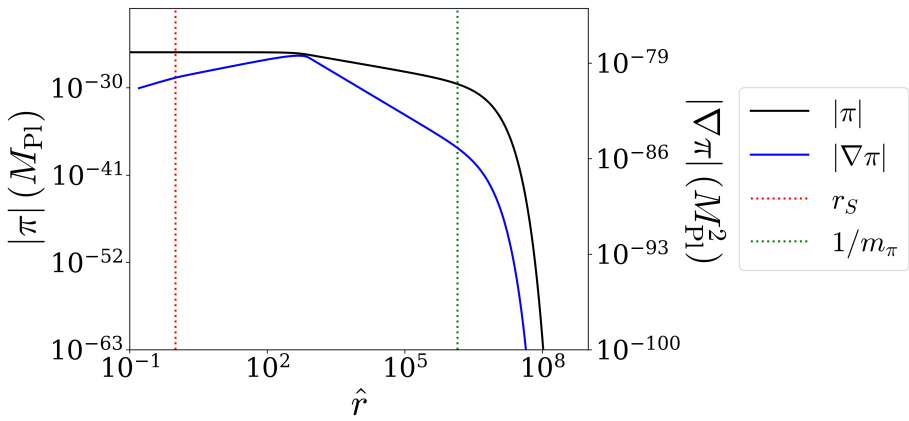

Figure 1: Plot of the field magnitude $|\pi|$ (black) and magnitude of field gradient $|\nabla\pi|$ (blue) across the box of size $10^9\,\hat{r}$, where $\hat{r} \equiv r/r_S$. Vertical lines are plotted for r_S (source radius) and $1/m_\pi$ (Compton wavelength). Parameters are given in Table 1. Note that the field gradient is negative throughout.

The gradient is proportional to the fifth force that arises from the π field, and to make apparent the behaviour of the force we present Figure 2. When the ratio of the force mediated by the π field and the standard force $F_\pi/F_N = 2$, it indicates that the forces are of equal strength (due to normalisation). For Vainshtein screening then, we expect the ratio < 2. Outside the Vainshtein radius $r_{\rm V}$, F_π is no longer screened and the ratio is $\mathcal{O}(1)$. However, at $r > m_\pi^{-1}$, we see that once again the ratio falls. We can interpret this as the force decaying when the π field reaches its background configuration. This finite force range arises due to the non-zero mass m_π, in contrast to General Relativity, where the Newtonian force scales as $1/r^2$ without exponential suppression.

4.2 Higher-derivative operators

Having reproduced the known $n = 3$ results and established the expected Vainshtein screening behaviour, we now turn to the question of whether this IR theory admits a controlled effective field theory interpretation. In particular, if the non-linear derivative interactions arise as part of an EFT expansion, higher-dimension operators should be progressively suppressed. This motivates an exploration of the impact of additional higher-derivative terms beyond the leading $n = 3$ interaction considered in [22].

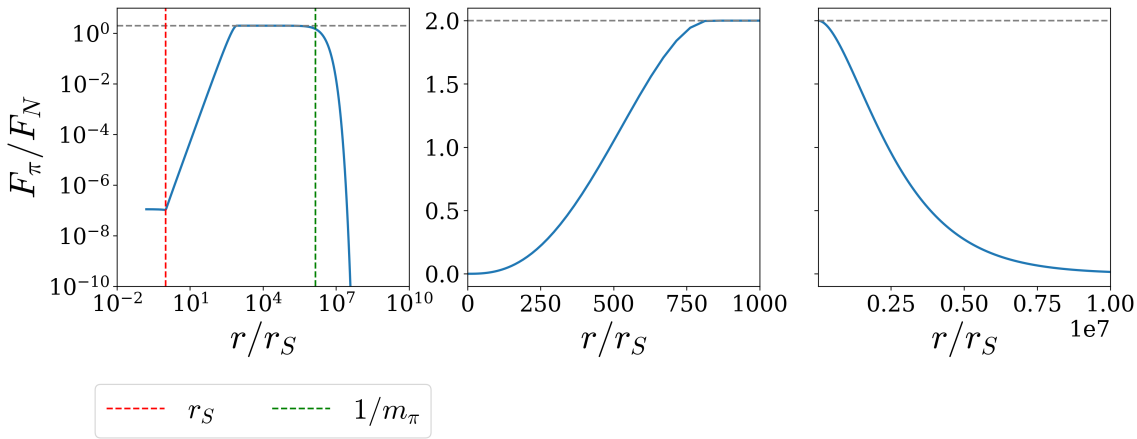

Figure 2: Plots of the ratio of the scalar force F_π and the standard Newtonian force F_N. A value of 2 corresponds to Vainshtein screening. *Left*: Ratio over the simulation box. Vertical lines are plotted for r_S (source radius) and $1/m_\pi$ (Compton wavelength). *Middle*: ratio around the source. *Right*: ratio shortly beyond m_π^{-1}. Note that the middle and right plots share a vertical axis scale. Parameters are given in Table 1.

To probe the robustness of the EFT picture, we therefore attempted to extend the model by including the next higher-order operator, solving the equation

$$\nabla_r^2\pi - m_\pi^2\pi - \epsilon\frac{\nabla_r^2\left((\nabla_r^2\pi)^3\right)}{\Lambda^8} - \epsilon\frac{\nabla_r^2\left((\nabla_r^2\pi)^4\right)}{\Lambda^{11}} = \frac{\rho}{M_{\rm Pl}}\,. \tag{4.2}$$

Our expectation was that the $n = 4$ contribution would remain sub-dominant across the domain of interest. While showing this would not guarantee a valid EFT, failure to exhibit such suppression would signal a breakdown of the effective description.

Initial attempts focused on implementing an arbitrary number of higher-derivative operators in a general and extensible manner. While this implementation correctly reproduced the $n = 3$ results when higher-order terms were disabled, including both the $n = 3$ and $n = 4$ operators led to rapidly growing residuals, indicating a failure of convergence. A subsequent hard-coded implementation of the combined $n = 3$ and $n = 4$ system exhibited similar behaviour.

Some sensitivity to solver stability was observed when varying parameters, scalings, and initial guesses, such as switching between non-linear and Klein-Gordon initial conditions [22]. However, even in cases where convergence was initially achieved, the solutions were not stable under extended iteration.

A persistent feature of these runs was the appearance of numerical noise in the region where the Vainshtein mechanism is expected to operate. We speculate that this may be associated with the competition between the $n = 3$ and $n = 4$ operators for dominance in this region, compounded by the presence of widely separated scales in the model.

To investigate whether this behaviour reflects a genuine physical effect or a limitation of the numerical method, we followed the suggestion of Professor Englert and introduced independent coefficients α and β controlling the strength of the non-linear operators. Schematically, we consider

$$(\Box + m^2)\phi + \alpha\mathcal{D}^3\phi + \beta\mathcal{D}^4\phi = \rho\,, \tag{4.3}$$

where $\mathcal{D}^n$ denotes the non-linear operator of order n. This allows β to be smoothly varied from zero, where the $n = 3$ limit is recovered.

While the $\beta = 0$ case reproduces the expected $n = 3$ solution, we find that even very small values of β (e.g. $\mathcal{O}(10^{-6})$) can lead to significant deformations of the field profile. For sufficiently small β, no discernible impact is observed, suggesting that the solver effectively neglects the higher-order term below some numerical threshold. Once the $n = 4$ contribution is included, however, the numerical solution deteriorates rapidly.

Further exploration of the (α, β) parameter space, as well as attempts to study models with all scales taken to be $\mathcal{O}(1)$, did not resolve these issues. As such, we are unable to draw firm conclusions regarding the physical role of higher-derivative operators in this model. While the observed behaviour may reflect a genuine breakdown of the EFT description in the Vainshtein regime, we suspect that the issue was our numerical implementation when dealing with competing non-linearities.

4.3 $n = 3$ and $n = 4$ case

Since stable numerical solutions to the combined $n = 3$ and $n = 4$ system could not be obtained, we instead probe the behaviour of the higher-dimension operators independently. This allows us to assess whether the qualitative features of the solution are sensitive to the operator dimension, and whether the effective field theory hierarchy is respected.

To this end, we examine how the solution depends on the EFT cutoff Λ. In Figure 3, we show the $n = 3$ solution with parameters given in Table 1, alongside $n = 4$ solutions with identical parameters except for Λ, which is varied over the range $10^{-47} \to 10^{-40}$. The $n = 3$ case corresponds to $\Lambda = 10^{-46}$.

Remarkably, the overall shape of the field profile is very similar across all cases. In particular, the $n = 3$ solution lies between the $n = 4$ solutions with $\Lambda = 10^{-46}$ and $\Lambda = 10^{-45}$. Increasing Λ systematically shifts the Vainshtein radius $r_{\rm V}$ closer to the source radius r_S, consistent with the expectation that larger cutoff scales suppress non-linear effects. For sufficiently large Λ (e.g. $\Lambda = 10^{-40}$), the flat region in the log-log plot disappears entirely, indicating the absence of Vainshtein screening. At large radii, all solutions converge prior to the Compton wavelength m_π^{-1}.

In Figure 4, we plot the log of the ratio of gradients of each $n = 4$ solution to the gradients of the $n = 3$ solution. Once again, the $n = 3$ profile lies between the

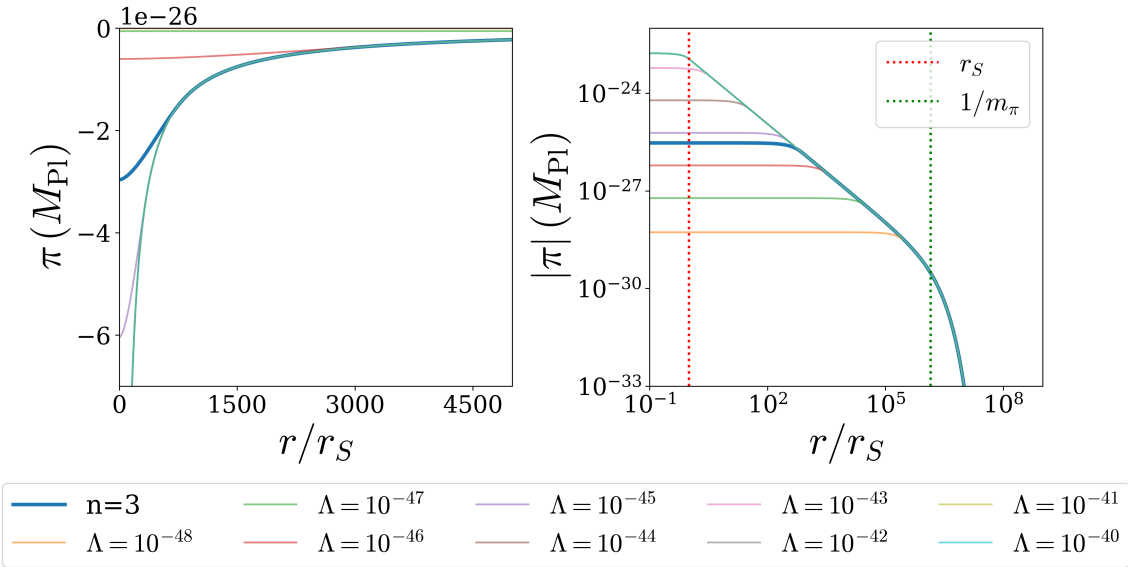

Figure 3: Plots of the field profile of the $n = 3$ case (with parameters given in Table 1) together with the field profiles of $n = 4$ solutions with the same parameters bar Λ, which is plotted for each value in the range $10^{-47} \to 10^{-40}$ inclusive. *Left*: π profiles around the source with linear axes. *Right*: $|\pi|$ profiles over the simulation box with log-log axes. Vertical lines are plotted for r_S (source radius) and $1/m_\pi$ (Compton wavelength).

$\Lambda = 10^{-46}$ and $\Lambda = 10^{-45}$ $n = 4$ cases. At large distances the solutions converge, as expected once the fields approach their Yukawa regime.

We find that for larger values of Λ the logarithm of the ratio becomes positive, indicating that the $n = 4$ theory mediates a stronger scalar force than the $n = 3$ theory, while for smaller Λ the ratio is negative. This is the opposite behaviour one expects for an EFT based purely on operator suppression by powers of Λ. In the Vainshtein region, the solutions for different Λ allow higher-derivative operators to induce larger gradients despite their higher dimensionality. This could suggest that the derivative expansion is not well behaved for screened backgrounds, and that higher-dimension operators can compete with or dominate lower-dimension ones.

5 Discussion & Outlook

In this report we motivated the modification of General Relativity through the introduction of an additional scalar degree of freedom, and investigated the resulting phenomenology. The theory of galileons was briefly reviewed, as these models are known to exhibit the Vainshtein mechanism, which allows modifications of gravity to evade local tests on solar-system scales [18].

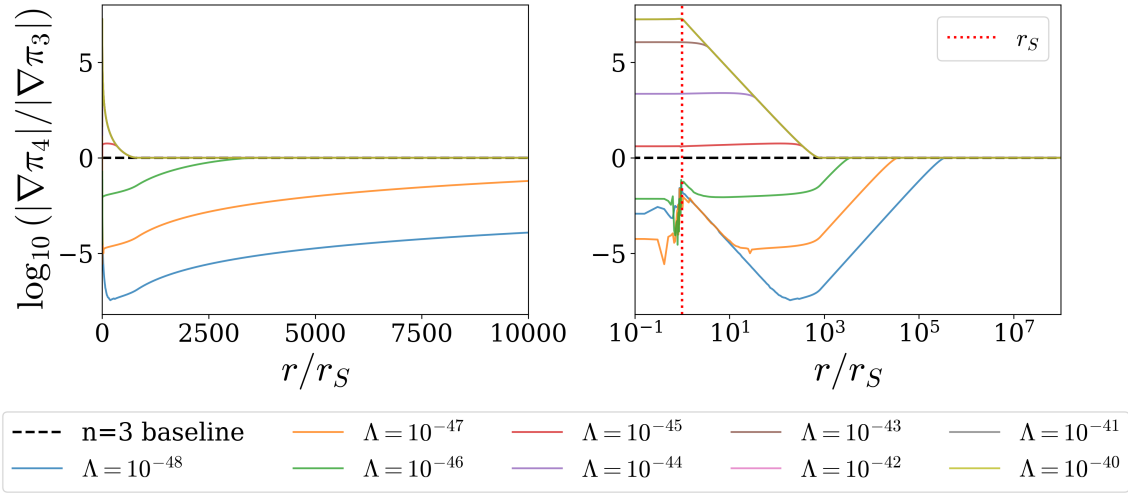

Figure 4: Plots of the log of the ratio of the $n = 4$ and $n = 3$ gradient terms. The $n = 3$ field has parameters given in Table 1, and the $n = 4$ fields have the same parameters but with Λ in the range $10^{-47} \to 10^{-40}$ inclusive. *Left*: ratio around source in linear-linear axes. *Right*: ratio over the simulation box with linear-log axes. A vertical line is plotted for r_S (source radius).

Building on the work of the Nottingham group [22, 44], we made use of and extended the current state-of-the-art numerical software φenics to study an infra-red galileon theory. This included investigating how Vainshtein screening appears via non-linear terms and probing the health of the effective field theory description of this IR model.

While our results do not allow for a definitive statement on the validity of the EFT, they provide indications that the expansion may not be well behaved in the Vainshtein regime. In particular, higher-derivative operators were found to compete with, and in some cases dominate over, lower-dimension operators despite suppression by the cutoff scale. This suggests that the expected EFT hierarchy is questionable.

An important limitation of this work is the inability to obtain stable numerical solutions when multiple higher-derivative operators are simultaneously included in the equations of motion. Future progress would benefit from a successful implementation of a numerical solver that can handle the competing non-linear terms. We believe this to be possible, although a more careful consideration of the implementation of the numerical methods would be required.

Another natural extension of this work would be to relax the assumptions of a static and spherically symmetric configuration. As discussed in [22], the finite element method implemented via `FEniCS` in φenics is well suited to tackling more general geometries. This could allow the study of screening in less idealised and more astrophysically relevant scenarios.

Finally, we note that much of the analysis and interpretation in this work is

based on numerical methods due to the highly non-linear nature of the equations considered here. In some regards, this is inevitable given the Vainshtein mechanism is by definition non-linear, however I believe that there is scope for a more detailed analytical analysis of these systems.

Acknowledgments

The work carried out in this project was inspired by the φenics paper [22] and uses the software [42]. The formatting of this report is primarily based upon the JHEP LaTeX format, with some minor modifications; the original files can be accessed here. The author would also like to recognise the helpful insight provided by both Christoph Englert and Peter Millington in their supervision of this project.

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